

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Providing a summary of the problem found in the DNS and ICMP traffic log.

The network traffic analysis indicates that DNS requests sent over UDP to **port 53** did not receive a valid response. Instead, the ICMP traffic returned a **Destination Unreachable – Port Unreachable** message. Since port 53 is used by the DNS service, this indicates that the DNS server could not accept or process the request.

The most likely cause of the issue is that the DNS service is unavailable, the DNS server is misconfigured, or a firewall is blocking traffic to port 53.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred at approximately **1:27 PM**, when users were unable to access the website because the domain name could not be resolved.

The IT team became aware of the issue after users reported that they could not access the website. The team used the **tcpdump** network protocol analyzer to capture and analyze the DNS and ICMP traffic.

This indicates that the DNS service on the destination server was unavailable or that network devices, such as a firewall, were blocking access to port 53.

A likely cause of the incident is a firewall configuration blocking DNS traffic, a DNS server outage, or a misconfiguration that prevented the DNS service from responding to requests.